Solution Requirements

CrimeVision AI — Intelligent Multimodal Forensic Analysis Platform

Date	29 October 2026	**Team ID**	Not applicable — solo project
Project Name	CrimeVision AI	**Maximum Marks**	4 Marks (template criterion)

Purpose and scope

This document defines the proposed functional and non-functional requirements for CrimeVision AI, a multimodal forensic analysis prototype. The system is intended to help an authorized investigator organize submitted case material, extract useful information from supported image/video, audio, and document evidence, correlate candidate findings, and review source-linked outputs. It supports investigation; it does not determine guilt or replace expert or legal judgment.

Scope assumptions

- Individual academic prototype; no team roles or team ID are applicable.
- Evidence is supplied by an authorized user in a controlled demonstration environment.
- Similarity matching, if demonstrated, uses a mock case database only—not a live law-enforcement or biometric database.
- AI-generated results are candidate leads and must be reviewed by a human.
- Requirements below describe intended behavior; targets are acceptance goals for the prototype, not claims of completed testing.

Step 1: Functional Requirements (FR)

Functional requirements describe what the system must do. Priorities are assigned for a feasible academic prototype.

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	Provide a controlled sign-in or equivalent access gate for the investigator interface. For a local demo, access may be limited to a configured user; do not claim production-grade identity assurance unless implemented.	High
2	Authorization levels	Restrict evidence upload, case viewing, analysis execution, and report access to authorized users. The prototype may use a single investigator role; role separation is optional unless multiple roles are implemented.	High
3	External interfaces	Integrate the application with the selected processing components: Flask API, OpenCV and RT-DETR/image-captioning for visual inputs, Whisper for audio transcription, OCR/text extraction for documents, and a Groq-hosted LLM for evidence-grounded synthesis where configured.	High
4	Transactions processing	Validate uploaded file type and size, create a case/evidence record, route each supported file to the appropriate analysis pipeline, track processing status, and return clear success or error states. Preserve the original input separately from derived outputs.	High
5	Reporting	Present a case dashboard with uploaded evidence, extracted observations, timestamps where available, source-file references, candidate correlations, and a downloadable or viewable summary report.	High
6	Business rules / workflow	Process evidence in stages: intake and validation; modality routing; modality-specific extraction; unified evidence record creation; cross-evidence correlation; human review and report generation. Label AI inferences separately from directly extracted facts.	High
7	Compliance to laws or regulations	Use only evidence the operator is authorized to process. Document consent/authority and retention expectations for any real data. Avoid unapproved personal-data collection; do not connect to live police, face-recognition, or restricted databases in the prototype.	High
8	Other	Maintain evidence provenance by associating derived results with the source file and available timestamp/page/frame reference. Provide processing errors and model/service unavailability messages; allow the investigator to review findings before report finalization.	High

Step 2: Non-Functional Requirements (NFR)

Non-functional requirements define how well the prototype should operate. Metrics are realistic proposed acceptance criteria for a local academic demonstration; they should be validated on the target hardware and representative files.

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	For a small demonstration file and a warm local application, the upload/status interface should respond promptly; analysis duration depends on model size, hardware, external API latency, and input length.	UI response ≤ 2 seconds for common actions; show processing status for longer jobs. Record actual per-modality processing times during testing.
2	Scalability	Support sequential processing of a small case set and multiple evidence items without mixing case records. The prototype is not designed for large-scale concurrent forensic workloads.	Demonstrate at least 5 evidence items in one case and 3 sequential cases without cross-case data leakage, subject to available storage and model limits.
3	Security & Data Privacy	Validate file extensions and content where possible; enforce upload limits; keep API keys in environment variables; avoid logging secrets; restrict access to stored evidence; and clearly separate original files from generated outputs.	No API keys in source control; reject unsupported/oversized files with clear errors; case data is not publicly exposed by default; document any external API data transfer.
4	Reliability & Availability	Handle malformed files, model/API failures, missing dependencies, and timeouts without crashing the entire application. Preserve completed results and indicate incomplete analysis.	A failed modality produces a visible error/status while other completed results remain accessible; no silent success when analysis fails.
5	Usability & Accessibility	Provide a clear upload workflow, case status, readable evidence summaries, source references, and explicit warnings that AI results require human verification.	A first-time tester can upload a supported file and locate its result without developer intervention; labels and errors are readable and keyboard-accessible where supported.
6	Other — Maintainability & Portability	Separate upload/API, modality processing, evidence correlation, persistence, and reporting into understandable modules. Document setup, model dependencies, configuration, and limitations.	A fresh development environment can be configured from documented steps; model/API settings are configurable without editing application logic.

Acceptance and verification

Area	Verification approach
Functional flow	Test supported image/video, audio, and document examples; verify routing, extracted outputs, source references, and report generation.
Failure handling	Test unsupported file types, empty/corrupt files, unavailable model/API services, and timeouts; confirm useful error messages.
Privacy & provenance	Verify original evidence remains unchanged, generated outputs identify their source, secrets are excluded from logs, and mock records are clearly labelled.
Human review	Confirm reports identify AI-derived candidate findings and do not present them as proof of identity, guilt, or definitive forensic conclusions.

Important: This document describes proposed requirements for the CrimeVision AI prototype. It does not certify that every feature is implemented, validated, legally compliant, or suitable for operational law-enforcement use. Final dates, measured performance, and implementation status should be confirmed during the project's final review.